Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/28/2005
PI Name:	Faeth, Gerard Michael Ph.D.		
Project Title:	Chemically Passive Suppression of Premixed Flames in Spacecraft Environments at Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Special Category:	None		
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Zip Code:	48109-2140	**Congressional District:**	15
Comments:	Faeth, Gerard M.--DECEASED Jan 2005 (per N. Hall/GRC)		
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Key Personnel Changes/Previous PI:	Faeth, Gerard M.--DECEASED Jan 2005 (per N. Hall/GRC)		
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This research investigates the effects of chemically-passive flame suppressants on the properties of laminar premixed flames at normal earth and microgravity conditions: considering various simple fuels for computational tractability with properties exhibiting various preferential-diffusion stability conditions that are known to affect laminar premixed flames (hydrogen and methane that are unstable at lean conditions, ethane that exhibits neutral stability and propane that is stable at lean conditions); various oxygen/nitrogen mixtures, fuel-equivalence ratios and pressures (to simulate spacecraft environments); and various chemically-passive flame suppressants (argon, nitrogen, carbon dioxide, and water mist). The measurements will be reduced to find effects of test conditions on fundamental unstretched laminar burning velocities as well as simple concepts of flame suppression effectiveness such as minimum oxygen indices, laminar burning velocities, flame temperatures, etc.

The objectives of this research are: (1) to measure the effect of typical chemically-passive flame suppressants on laminar

Task Description:	premixed flames at microgravity conditions relevant to spacecraft, and (2) to use the measurements to evaluate both detailed numerical simulations of the flames as well as simplified concepts of flame suppressant effectiveness such as minimum oxygen indices, laminar burning velocities, flame temperatures, etc., in order to provide rational rules for developing practical flame suppressants for spacecraft. The significance of this research stems from the justification for limiting the study to consideration of effects of chemically-passive suppressants on the properties of laminar premixed flames. First of all, only chemically-passive suppressants are considered because chemically-active suppressants, such as Halons, generally are not allowable for spacecraft applications due to problems of astronaut health concerns and potential poisoning of spacecraft life support systems. Next, although most practical flames are turbulent, turbulent flames are difficult to study because experimental conditions are substantially complicated by the need to describe turbulence properties. In addition, turbulent flames cause turbulence-induced stretching and quenching, making them less conservative indicators of flame extinction than corresponding laminar flames. Another advantage of laminar flames is that they generally are tractable for detailed numerical simulations, unlike turbulent flames, enhancing capabilities to use computations to supplement directly measured properties. Laminar flames also are relevant to practical turbulent flames based on widely-accepted laminar flamelet concepts of turbulent flames. Finally, due to the complexities of turbulent flames, it seems unlikely that understanding of turbulent flame suppression will precede understanding of laminar flame suppression. Thus, for these reasons, it seems prudent to consider the suppression of laminar flames at the outset.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research benefits both the advancement of fundamental combustion science and the terrestrial fire suppression engineering community and industry.
Task Progress:	Work during this report period involved experimental and computational evaluation of the effects of chemically-passive suppressants on laminar premixed hydrogen/oxygen/nitrogen/ suppressant flames at conditions where laminar flame speeds are reasonably fast so that effects of earth gravity are small; and the design/development of the short-drop free-fall apparatus for experiments to find flammability limits for various suppressants when effects of earth gravity are invariably important. The studies of hydrogen/oxygen/nitrogen/suppressant flames involved experiments and numerical simulations for the following conditions: premixed hydrogen/air suppressant and hydrogen/30% oxygen and 70% nitrogen (by volume)/suppressant (the EVA-preparation atmosphere) flames; normal temperature (300 K); fuel-equivalence ratios of 1.0 and 1.8; pressures of 0.5, 0.7 and 1.0 atm; gaseous flooding type suppressants including helium, argon, nitrogen and carbon dioxide; and suppressant concentrations of 0-40% (by volume) which implies oxygen indices of 30-10. Major findings were that the suppressants increased in effectiveness in the order helium, argon, nitrogen and carbon dioxide, which reflects their progressively increased capabilities to either quench the reaction zone by their increased specific heats or to reduce flame velocities due to their reduced transport rates; the addition of chemically-passive suppressants generally decrease flame Markstein numbers which made them more susceptible to preferential-diffusion instability which increases flame speeds and counteracts effects of suppressants; hydrogen-fueled flames are unusually robust and still exhibit rapidly-propagating flames at oxygen indices smaller than ten; and predictions and measurements were in excellent agreement. With the study of hydrogen/oxygen/nitrogen/suppressant flames completed, current work is concentrating on the development of a short-drop free fall facility needed for measurements of the effect of chemically-passive suppressants on flammability limits. Completing the development of this apparatus and initiating tests to find flammability limits for typical fuels, ambient atmospheres and passive suppressants will be the main focus of work during the next report period.
Bibliography Type:	Description: (Last Updated: 10/19/2009)
Articles in Peer-reviewed Journals	Qiao, L.; Kim, C. H.; Faeth, G. M. "Effects of Chemically-Passive Suppressants on Laminar Premixed Hydrogen/Air Flames" Combustion and Flame , Jan-2005
Awards	Faeth, G. M. "National Associate, National Academies" Jan-2004
Awards	Faeth, G. M. "Space Processing Award, American Institute of Aeronautics and Astronautics" Jan-2004
Awards	Faeth, G. M. "Alfred C. Egerton Gold Medal, The Combustion Institute" Jul-2004
Awards	Faeth, G. M. "Arthur B. Modine Distinguished University Professor of Aerospace Engineering, The University of Michigan" Sep-2004
Presentation	Qiao, L.; Kim, C. H.; Faeth, G. M. "Effects of Chemically-Passive Suppressants on Laminar Premixed Hydrogen/Air Flames" Workshop on Strategic Research to Enable NASA's Exploration Missions, Cleveland, OH Jun-2004
Presentation	Qiao, L.; Kim, C. H.; Faeth, G. M. "Effects of Chemically-Passive Suppressants on Laminar Premixed Hydrogen/Air Flames" Thirtieth International Symposium on Combustion, Chicago, IL Jul-2004